3896. Minimum Operations to Transform Array into Alternating Prime

Solved

Medium Hint

You are given an integer array `nums`.

An array is considered **alternating prime** if:

- Elements at **even** indices (0-based) are **prime** numbers.
- Elements at **odd** indices are **non-prime** numbers.

In one operation, you may **increment** any element by 1.

Return the **minimum** number of operations required to transform `nums` into an **alternating prime** array.

A **prime** number is a natural number greater than 1 with only two factors, 1 and itself.

Example 1:

> **Input:** nums = [1,2,3,4]
>
> **Output:** 3
>
> **Explanation:**
>
> - The element at index 0 must be prime. Increment `nums[0] = 1` to 2, using 1 operation.
> - The element at index 1 must be non-prime. Increment `nums[1] = 2` to 4, using 2 operations.
> - The element at index 2 is already prime.
> - The element at index 3 is already non-prime.
>
> Total operations = `1 + 2 = 3`.

Example 2:

> **Input:** nums = [5,6,7,8]
>
> **Output:** 0
>
> **Explanation:**
>
> - The elements at indices 0 and 2 are already prime.
> - The elements at indices 1 and 3 are already non-prime.
>
> No operations are needed.

Example 3:

> **Input:** nums = [4,4]
>
> **Output:** 1
>
> **Explanation:**
>
> - The element at index 0 must be prime. Increment `nums[0] = 4` to 5, using 1 operation.
> - The element at index 1 is already non-prime.
>
> Total operations = 1.

Constraints:

- $1 <= nums.length <= 10^5$
- $1 <= nums[i] <= 10^5$

Seen this question in a real interview before? 1/6

Yes No

Accepted 30,668/57.9K | Acceptance Rate 53.0%

Topics

Hint 1

Hint 2

Hint 3

Hint 4

Hint 5

Discussion (25)

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